

D416

DevNet Fundamentals

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A network engineer, who manages 300 switches for a school district, wants a script to add a brand-new VLAN resource to a controller through its REST API. The district recently bought new laptops for teachers. Which method should the script use?

- A. POST
- B. DELETE
- C. PATCH
- D. GET

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A script sends a request that changes only the description field of an existing interface, leaving other fields alone. Which method fits best?

- A. POST
- B. PATCH
- C. GET
- D. PUT

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A script at a logistics company, which runs three shifts a day, sends a valid token to an API and receives a 403 response. What does this mean?

- A. The token is missing or invalid.
- B. The caller is authenticated but not allowed to do this action.
- C. The resource does not exist.
- D. The server crashed.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

An API returns 429 after a script sends 500 requests in one minute. What should the script do?

- A. Retry immediately in a tight loop.
- B. Get a new token.
- C. Slow down and retry after the wait time the server gives.
- D. Report a server crash.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

An API for a hospital network, which has a new CIO, returns JSON that Python loads as `data = {"device": "edge1", "interfaces": [{"name": "Gi0/1"}, {"name": "Gi0/2"}]}`. Which expression returns "Gi0/2"?

- A. `data["interfaces"][2]["name"]`
- B. `data["name"]["interfaces"][1]`
- C. `data[1]["name"]`
- D. `data["interfaces"][1]["name"]`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Given the same data, what does `data["device"]` return?

- A. The list of interfaces.
- B. A `KeyError`.
- C. "Gi0/1"
- D. "edge1"

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A developer, who works remotely from a small town, ran `git add` and `git commit` on a new script. Her teammate cannot see the change in the shared repository. What step is missing?

- A. `git push`
- B. `git add`
- C. Nothing; commit already shares the change.
- D. `git pull`

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A teammate pushed fixes to the remote main branch. How does a developer get those changes into her local copy?

- A.** git commit
- B.** git push
- C.** git pull
- D.** git init

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

Two engineers on a team that just finished a big migration both edit line 12 of the same config template on different branches. When the second branch is merged, what happens?

- A.** Git keeps the newest edit automatically.
- B.** The first branch is deleted.
- C.** Both edits are silently combined on one line.
- D.** A merge conflict that must be resolved by hand.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A script that pulls data from a controller has the API token typed into the code. The script is about to be committed to a shared Git repository used by 40 engineers across the company's four regional offices. What should be done?

- A.** Commit it; only employees can see the repository.
- B.** Share it only with the four offices.
- C.** Read the token from an environment variable or vault, and keep it out of the code.
- D.** Put the token in a code comment instead.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A script logs in to a REST API, which is hosted by a large cloud provider, and gets a token. Its next request leaves the token out and receives 401. Why?

- A.** Each REST request must carry its own authentication; the server keeps no session.
- B.** The user lacks permission.
- C.** The server remembers the login, so the 401 is a bug.
- D.** The cloud provider is having an outage.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A carrier’s automation team, which recently hired three new engineers, uses structured data models to define interface settings and sends them to routers as XML over SSH. What are the model language and protocol?

- A. NETCONF models over YANG.
- B. YANG models over NETCONF.
- C. YANG models over RESTCONF.
- D. SNMP traps.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A network engineer, ~~who manages 300 switches for a school district,~~ wants a script to add a brand-new VLAN resource to a controller through its REST API. ~~The district recently bought new laptops for teachers.~~ Which method should the script use?

Target word: method to create **Decides:** Creating a new resource.

A	Correct	POST creates a new resource.
B	Confusion set	Removes a resource.
C	Adjacent category	Partly updates an existing resource.
D	Inverted operation	GET only reads.

2. Answer B Concept 1 · transfer

A script sends a request that changes only the description field of an existing interface, leaving other fields alone. Which method fits best?

Target word: method for partial update **Decides:** Existing resource, only one field.

A	Confusion set	Creates a new resource.
B	Correct	Partial update.
C	Inverted operation	Reads only.
D	Adjacent category	Replaces the whole resource.

3. Answer B Concept 2

A script at a logistics company, ~~which runs three shifts a day,~~ sends a valid token to an API and receives a 403 response. What does this mean?

Target word: meaning of 403 **Decides:** Token is valid but access is denied.

A	Confusion set	That is 401.
B	Correct	403 is forbidden.
C	Adjacent category	That is 404.
D	Inverted operation	Server errors are 5xx.

4. Answer C Concept 2 · transfer

An API returns 429 after a script sends 500 requests in one minute. What should the script do?

Target word: response to 429 **Decides:** Too many requests.

A	Inverted operation	Makes the limit worse.
B	Confusion set	Authentication is fine.
C	Correct	429 signals rate limiting.
D	Adjacent category	Not a 5xx.

5. Answer D Concept 3

An API for a hospital network, ~~which has a new CIO,~~ returns JSON that Python loads as `data = {"device": "edge1", "interfaces": [{"name": "Gi0/1"}, {"name": "Gi0/2"}]}`. Which expression returns "Gi0/2"?

Target word: access a nested value **Decides:** interfaces is a list; Gi0/2 is at index 1.

A	Wrong stem number	Index 2 does not exist; counting starts at 0.
B	Inverted operation	Reverses the nesting order.
C	Dropped element	Skips the interfaces key; dictionaries are not indexed by number.
D	Correct	Key, then zero-based index, then key.

6. Answer D Concept 3 · transfer

Given the same data, what does `data["device"]` return?

Target word: value at a top-level key **Decides:** device maps to a string.

A	Confusion set	That is under the interfaces key.
B	Inverted operation	The key exists.
C	Adjacent category	A nested value, not the device name.
D	Correct	Direct key lookup.

7. Answer A Concept 4

A developer, ~~who works remotely from a small town,~~ ran `git add` and `git commit` on a new script. Her teammate cannot see the change in the shared repository. What step is missing?

Target word: missing Git step **Decides:** Commits are local until pushed.

A	Correct	Sends local commits to the remote.
B	Dropped element	Already done.
C	Absolute language	Commit is local only.
D	Inverted operation	Brings remote changes down, not up.

8. Answer C Concept 4 · transfer

A teammate pushed fixes to the remote main branch. How does a developer get those changes into her local copy?

Target word: command to receive changes **Decides:** Fetch and merge from remote.

A	Confusion set	Records local changes.
B	Inverted operation	Sends changes up.
C	Correct	Fetches and merges remote changes.
D	Adjacent category	Creates a new repository.

9. Answer D Concept 5

Two engineers on a team ~~that just finished a big migration~~ both edit line 12 of the same config template on different branches. When the second branch is merged, what happens?

Target word: merge result **Decides:** Same lines changed on both sides.

A	Absolute language	Git does not choose by time.
B	Adjacent category	Merging does not delete branches.
C	Inverted operation	Only different lines merge automatically.
D	Correct	Git cannot choose between them.

10. Answer C Concept 6

A script that pulls data from a controller has the API token typed into the code. The script is about to be committed to a shared Git repository used by 40 engineers ~~across the company's four regional offices~~. What should be done?

Target word: credential handling **Decides:** Secrets in code will be shared through Git.

A	Inverted operation	Anyone with access, now or later, gets the token.
B	Camouflage	Office count does not fix exposure.
C	Correct	Secrets stay out of version control.
D	Dropped element	Still in the repository.

11. Answer A Concept 7

A script logs in to a REST API, ~~which is hosted by a large cloud provider~~, and gets a token. Its next request leaves the token out and receives 401. Why?

Target word: why the second call failed **Decides:** REST is stateless.

A	Correct	Statelessness.
B	Confusion set	That is 403.
C	Inverted operation	Stateless servers do not remember.
D	Camouflage	That would be a 5xx.

12. Answer B Concept 8

A carrier's automation team, ~~which recently hired three new engineers~~, uses structured data models to define interface settings and sends them to routers as XML over SSH. What are the model language and protocol?

Target word: name model and protocol **Decides:** Data model plus XML over SSH.

A	Inverted operation	Swaps model and protocol.
B	Correct	NETCONF uses XML over SSH.
C	Confusion set	RESTCONF runs over HTTP.
D	Adjacent category	An older monitoring method.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	12	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	HTTP methods and CRUD	GET reads, POST creates, PUT replaces, PATCH partly updates, DELETE removes.	Using GET to change data, or POST and PUT interchangeably.	An API call that reads, creates, or changes a resource.
2	HTTP status codes	2xx success; 401 not authenticated; 403 authenticated but not allowed; 404 not found; 429 too many requests; 5xx server error.	Mixing up 401 and 403, or blaming the client for a 5xx.	A response code and a question about the cause.
3	Parsing JSON	JSON objects become Python dictionaries (access by key) and arrays become lists (access by zero-based index).	Indexing a dictionary by number or a list by key.	Nested JSON and a request for one value.
4	Git commit, push, and pull	git add stages; git commit records locally; git push sends commits to the remote; git pull fetches and merges remote changes.	Thinking commit uploads to the shared repository.	Work that is saved locally but teammates cannot see.
5	Branches and merge conflicts	Work on a branch and merge it back; a conflict happens when both sides changed the same lines and must be resolved by hand.	Thinking Git picks the newest change on its own.	Two people editing the same lines.
6	Handling API credentials	Send tokens in request headers and keep secrets out of code — use environment variables or a vault.	Hardcoding a token in a script committed to Git.	A key or password inside source code.
7	REST statelessness	Each REST request carries everything the server needs, including authentication; the server keeps no client session between calls.	Expecting the server to remember a login from an earlier request.	A second call failing without the token.
8	Model-driven programmability	YANG models define structured device data; NETCONF (XML over SSH) and RESTCONF (HTTP with JSON or XML) read and change it.	Calling YANG a transport protocol.	Structured config data vs. the protocol that carries it.

Course scope (public description): The DevNet Fundamentals course teaches students how to automate and deploy network solutions in the Cisco Environment. This course includes APIs, Scripting, Python programming, and software version control.